07. Model Training & Hyperparameter Tuning

Distinguish the experiments actually performed from a sound tuning workflow.

Project: Child Malnutrition Status Classification (ICFDR ML Project)

Source archive: ICFDR---ML-Project-main.zip | Main notebook: MalnutritionDetect.ipynb | Additional notebooks: SMOTE.ipynb and Oversampling on the whole dataset.ipynb

Main notebook experiments

- Random Forest with n_estimators=300; several cells train and evaluate on the same 70/15/15 split.
- Linear SVM with default settings; one branch encodes labels before fitting and another fits string labels directly.
- A dense neural network trains for 10 epochs with batch_size=32, but uses a binary output/loss for a four-class target.
- LabelEncoder is used for target labels.

SMOTE notebook experiments

- Split first into 585/126/126; SMOTE expands training data from 585 to 1,940 rows, with 485 examples per class.
- Random Forest with 100 trees; classification report and confusion matrix are printed for validation and test.
- Linear SVM is tried; then StandardScaler and GridSearchCV tune C and kernel using 3-fold CV.
- The reported best SVM parameters are C=10 and kernel=rbf.

Whole-dataset oversampling notebook

A separate notebook applies SMOTE to all 837 rows before the train/validation/test split. This creates a balanced dataset of 2,784 rows (696 per class), then splits those synthetic/real records. This is leakage: synthetic points may be constructed using neighbours that later appear in evaluation partitions.

Random Forest tuning

The whole-dataset notebook explores GridSearchCV with n_estimators, max_depth, min_samples_split, min_samples_leaf and bootstrap; it also contains RandomizedSearchCV with 100 iterations and 5-fold CV. Because this branch starts from globally SMOTE-resampled data, its validation and test estimates are not trustworthy.

Recommended tuning workflow

- Use an imbalanced-learn Pipeline with StandardScaler (for SVM) and SMOTE, ensuring SMOTE is applied separately inside each training fold.
- Use StratifiedKFold; with only four Wasting examples, ordinary CV will be unstable or impossible for every fold to contain that class.
- Tune scoring='f1_macro' or a carefully selected class-weighted objective, and inspect per-class recall.
- Compare class_weight='balanced' models against SMOTE; synthetic oversampling is not always superior.
- Record random seeds, package versions, best parameters and the untouched test result once.